Taichi Kamei

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Education

3rd Year Engineering Physics *Expected Graduation - May 2028*
University of British Columbia | Vancouver

Relevant Courses: Solid and Fluid Mechanics, Mechanical Designs, Electric Circuit Analysis, Signals and Systems, Thermodynamics, Statistical Mechanics, Quantum Mechanics

Dean's Honor List 2025

Experience

Firmware & Validation Engineer Co-op *May 2026 - Present*
UBC Blusson Quantum Matter Institute

- Developing full-stack embedded firmware for the multi-channel low-noise SNSPD biasing quasi-current source in C++, achieving lower noise spectral density than commercial SMU
- Designed an active object architecture on ESP32 with ESP-IDF, FreeRTOS, dual-core task isolation, inter-core queue communication, and event coordination for deterministic concurrent per-channel control
- Developed bidirectional V-I sweep measurement mode with hysteresis to characterize the switching current of the superconducting nanowire used for identifying a SNSPD steady biasing current
- Designed steady-state biasing mode with configurable voltage and duration for SNSPD operating point control
- Implementing 24-bit sigma-delta ADC and 16-bit DAC SPI driver from datasheet register maps
- Building PySide6 GUI and CSV data logger for user interface and data analysis
- Testing and verifying the noise floor, negative rail voltage noise, time domain voltage drift characteristics, and DAC & ADC accuracy of the instrument using oscilloscope and pico-ammeter
- Designing a PCB casing on SolidWorks, reducing vibrational noise coupling into the PCB

Projects

Drone Flight Controller *Feb 2026 - Present*

- Designed a power board for 4S LiPo, integrating BMS IC and 5V/5A buck converter
- Designed a custom 4-layer GPS-module & magnetometer PCB on KiCAD fitting within 260mm by 260mm
- Prototyping a flight controller PCB, integrating ESP32-S3-Mini-1U, BMS, IMU, barometer, and a RF transceiver using SX1261
- Planned to develop BMS I2C driver, GPS-module UART driver, and RF SPI firmware in Rust

Autonomous Payload Retrieving Robot *June - August 2025*

- Designed a 3-DOF four-bar-linkage arm and arm base on Onshape and fabricated components using 3D printer, water jet cutter, and laser cutter
- Sized appropriate servo motors for the arm from maximum payload and bending moment calculations
- Implemented magnetic encoder-based rotational control for the arm base, achieving accuracy of less than 1° error
- Designed a custom 2-layer I2C multiplexer and I2C buffer PCB on KiCAD to solve peripheral address conflicts and signal degradation
- Prototyped a payload detection algorithm in Python on Raspberry Pi using two 2D LiDAR, generating depth and reflectance map in 7Hz. Implemented it in C++ on ESP32 with 15 Hz real-time detection

Skills

Software	C/C++, Python, Java, Assembly, VHDL, Linux, Git
Embedded	ESP-IDF, FreeRTOS, Arduino, Raspberry Pi, FPGA, ROS, I2C, SPI, UART
Electrical	Kicad, Altium, LTSpice, Soldering, Oscilloscope, DMM, Spectrum Analyzer, Logic Analyzer
Mechanical	Onshape, Siemens NX, 3D printing, Laser Cutting, Water Jet Cutting, Drill Press, Caliper